MIT Linear Algebra Exam '22

Problem 1

a) any solution x to $Ax=b$ (if it exists) is always a sum of a vector in the Zeilenraum space of A plus a vector in the Nullraum space of

b) $Ax=b$ is solvable if (and only if) b is orthogonal to every vector in the Linken Nullraum space of A

($(C(A))^{\perp} = N(A^T)$)

c)